

HRG thermodynamics with Thermal-FIST

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An example on using open-source C++ code for scientific calculations

June 10, 2025



Hadron resonance gas (HRG)

HRG: Equation of state of hadronic matter as a multi-component (non-)interacting gas of known hadrons, resonances, and light nuclei (~ 400 components)

$$\ln Z \approx \sum_{i \in M, B} \ln Z_i^{id} = \sum_{i \in M, B} \frac{d_i V}{2\pi^2} \int_0^\infty \pm p^2 dp \ln \left[1 \pm \exp \left(\frac{\mu_i - E_i}{T} \right) \right]$$

Grand-canonical ensemble: $\mu_i = b_i\mu_B + q_i\mu_Q + s_i\mu_S$ *chemical equilibrium*

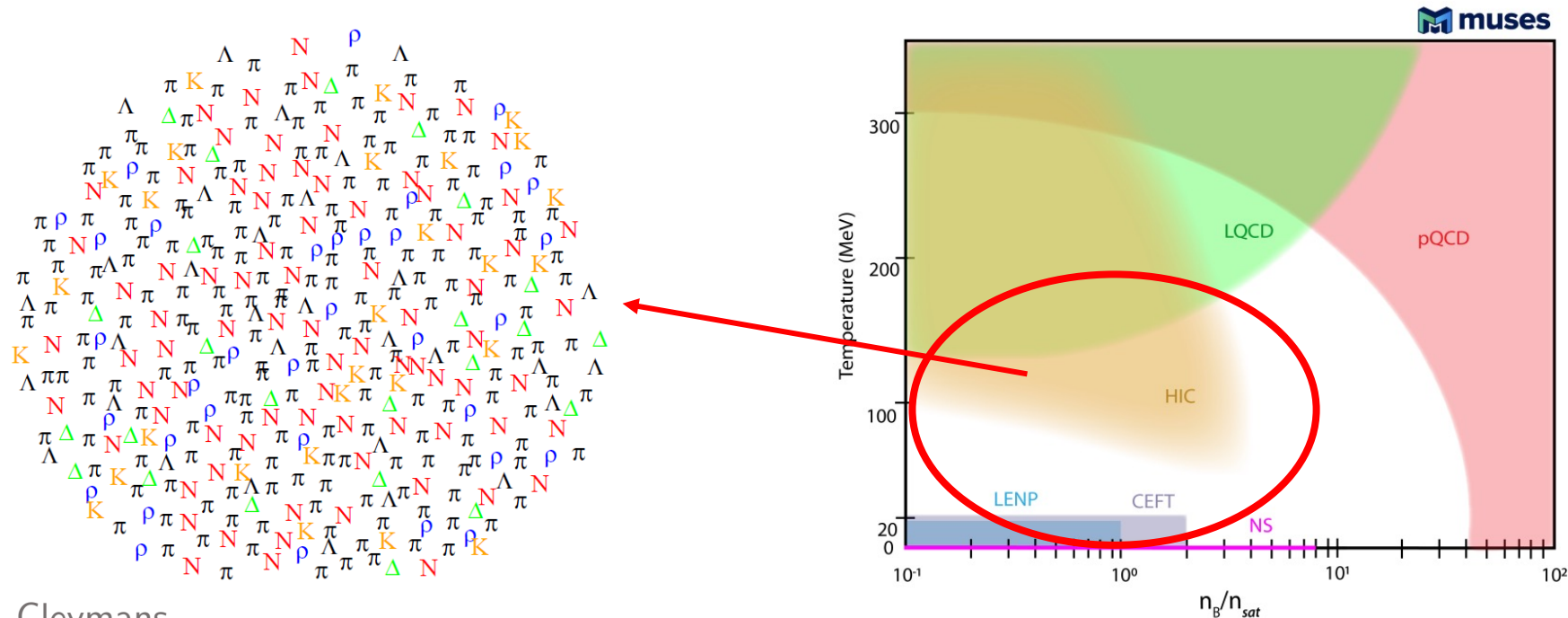


Figure from 2303.17021

Hadron resonance gas (HRG)

HRG describes the hadronic part of the QCD EoS (low temperature) and the hadron abundances measured in heavy-ion collisions

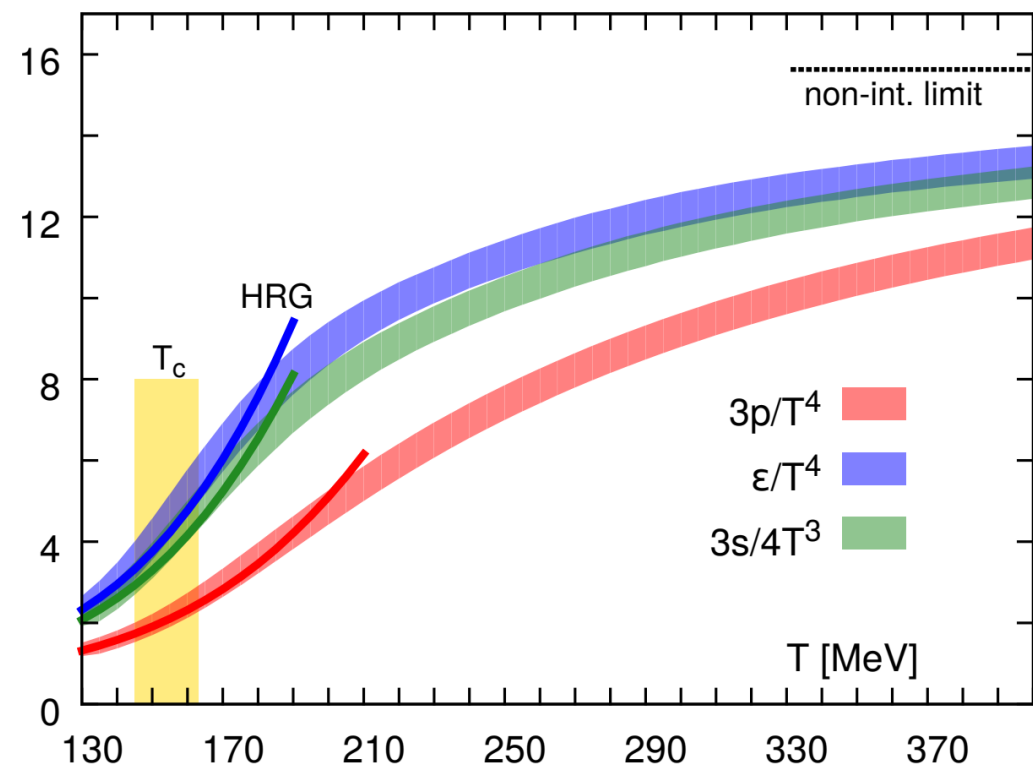
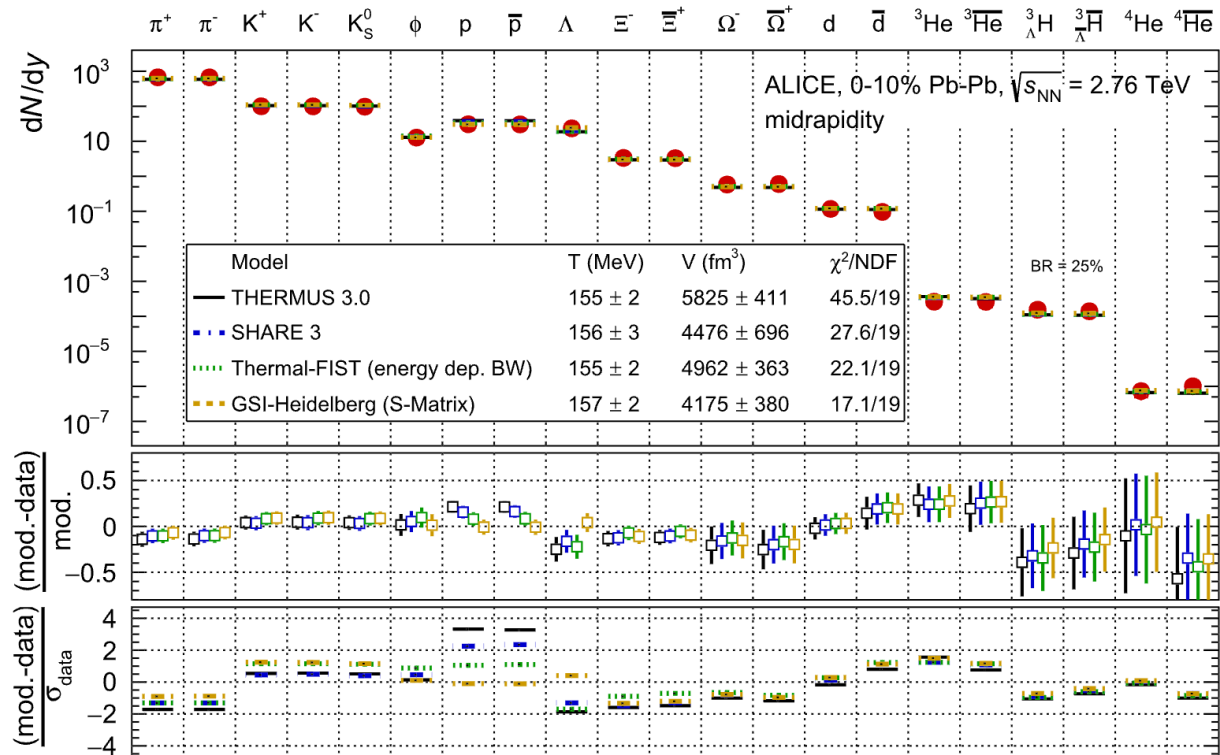


Figure from HotQCD coll., PRD '14



ALICE Collaboration, EPJC 84, 813 (2024)

What is Thermal-FIST?



Thermal-FIST* (current version: v1.5) [V.Vovchenko, H.Stoecker]

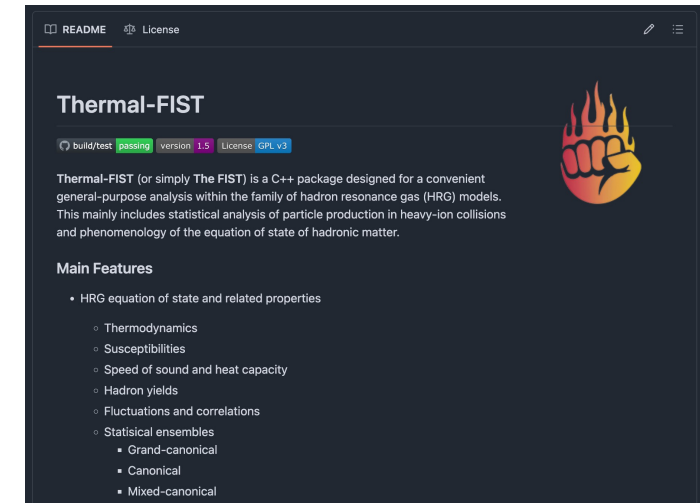
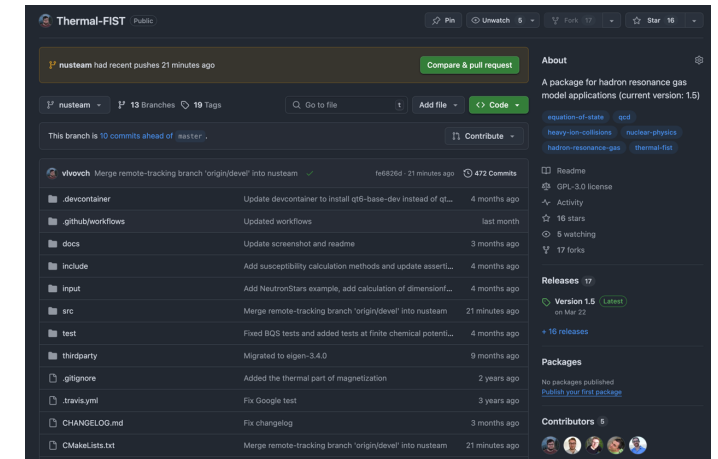
Open-source C++ package for general-purpose HRG model analysis
Cross-platform (Linux, Mac, Windows) through **cmake**

License: GPL-3.0

GitHub: <https://github.com/vlvovch/Thermal-FIST>

physics manual: [Comput. Phys. Commun. 244, 295 \(2019\)](#)

- **2014-2018:** Initial development and applications (closed source)
- **June 2018:** First public release (v0.6)
- **Jan 2019:** Code documentation and CPC article release (v1.0)
- **2019-2022:** Incremental upgrades
- **2025-...:** Version 1.5 with new features (dense matter EoS, cosmology)



Using Thermal-FIST

Installation using **git** and **cmake**

```
# Clone the repository from GitHub
git clone https://github.com/vlvovch/Thermal-FIST.git
cd Thermal-FIST

# Create a build directory, configure the project with cmake
# and build with make
mkdir build
cd build
cmake ../
make

# Run the GUI frontend
./bin/QtThermalFIST

# Run the test calculations from the paper
./bin/examples/cpc1HRGTDep
./bin/examples/cpc2chi2
./bin/examples/cpc3chi2NEQ
./bin/examples/cpc4mcHRG
```

Using Thermal-FIST: Console mode

```
#include "HRGBase.h"
#include "HRGEV.h"
#include "HRGFit.h"
#include "HRGVDW.h"

#include "ThermalFISTConfig.h"

using namespace std;

#ifdef ThermalFIST_USENAMESPACE
using namespace thermalfist;
#endif

// Temperature dependence of HRG thermodynamics at  $\mu = 0$ 
// Three variants of the HRG model:
// 1. Ideal HRG: <config> = 0
// 2. EV-HRG with constant radius parameter  $r = 0.3$  fm for all hadrons (as in 1412.5478): <config> = 1
// 3. QvdW-HRG with  $a$  and  $b$  for baryons only, fixed to nuclear ground state (as in 1609.03975): <config> = 2
// Usage: cpc1HRGTDep <config>
int main(int argc, char *argv[])
{
    // Particle list file
    // Here we will use the list from THERMUS-2.3, for comparing the results with THERMUS-2.3
    string listname = string(ThermalFIST_INPUT_FOLDER) + "/list/thermus23/list.dat";

    // Alternative: use the default PDG2014 list
    //string listname = string(ThermalFIST_INPUT_FOLDER) + "/list/PDG2014/list.dat";

    // Create the hadron list instance and read the list from file
    ThermalParticleSystem TPS(listname);

    // Which variant of the HRG model to use
    int config = 0;
```

```
if (config == 0) // Ideal HRG
{
    model = new ThermalModelIdeal(&TPS);

    printf("#Calculating thermodynamics at  $\mu = 0$  in Id-HRG model\n");

    modeltype = "Id-HRG";
}
else if (config == 1) // EV-HRG,  $r = 0.3$  fm, to reproduce 1412.5478
{
    model = new ThermalModelEVDiagonal(&TPS);
    // Set  $r = 0.3$  fm for each hadron in the list
    double rad = 0.3;
    for (int i = 0; i < model->TPS()->ComponentsNumber(); ++i)
        model->SetRadius(i, rad);

    printf("#Calculating thermodynamics at  $\mu = 0$  in EV-HRG model with  $r = %lf$  fm\n", rad);

    modeltype = "EV-HRG";
}
else if (config == 2) // QvdW-HRG, to reproduce 1609.03975
{
    model = new ThermalModelVDWFull(&TPS);

    // vdW parameters, for baryon-baryon, antibaryon-antibaryon ONLY, otherwise zero
    double a = 0.329; // In  $\text{GeV} \cdot \text{fm}^3$ 
    double b = 3.42; // In  $\text{fm}^3$ 
```

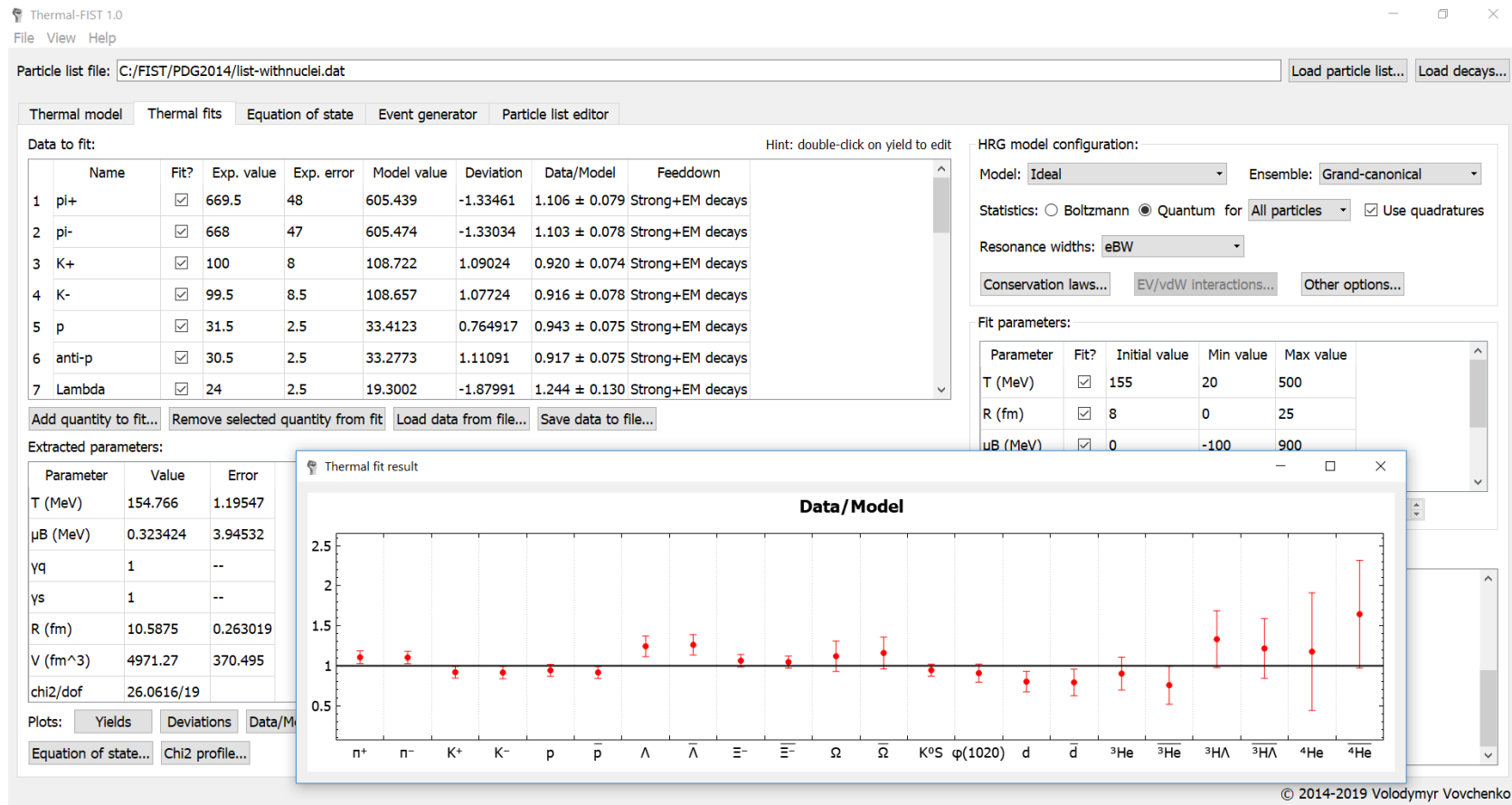
Link to **libThermalFIST** and write a C++ macro doing whatever calculation you want

The most flexible way of using the code

git submodule is useful

Using Thermal-FIST: GUI

Graphical user interface for *general-purpose* HRG model applications



Need to install free Qt6 framework from <https://www.qt.io/download>

Example case: HRG thermodynamics

Example case: HRG thermodynamics

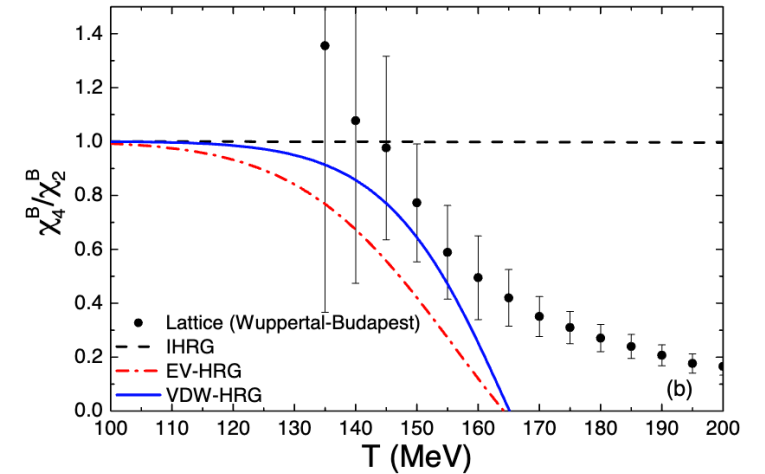
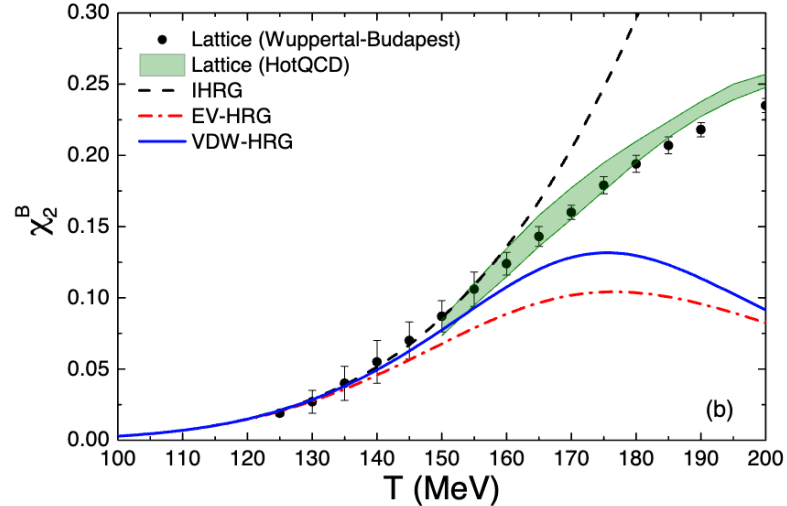
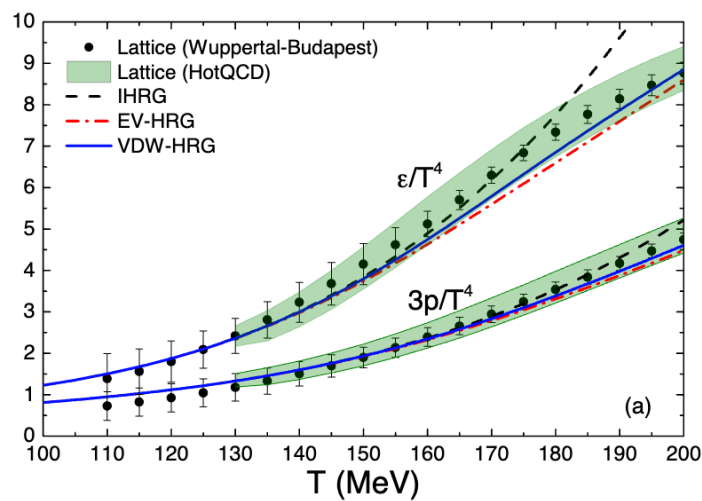
Go to <https://github.com/vlvovch/Thermal-FIST/tree/nusteam/src/nusteam> and follow the instructions

Necessary tools:

- **git** (source version control)
- **vscode** (preferred), Xcode, Clion, Cursor/Windsurf, or Visual Studio (Windows)
- **CMake** tool
 - On Windows use cmake-gui (<https://cmake.org/cmake/help/latest/manual/cmake-gui.1.html>) to generate a .sln project for use in Visual Studio
- **Qt6** framework for running graphical user interface <https://www.qt.io/download>
 - **Not required** for the above example but useful to play around with

Compare the results to published ones

V. Vovchenko, M.I. Gorenstein, H. Stoecker, *van der Waals Interactions in Hadron Resonance Gas: From Nuclear Matter to Lattice QCD*, Phys.Rev.Lett. 118 (2017) 182301, arXiv:1609.03975 [hep-ph]

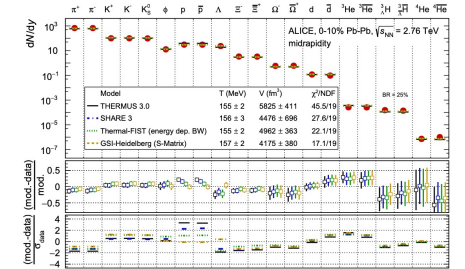
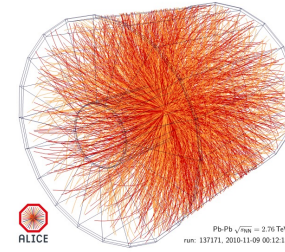


Extra slides

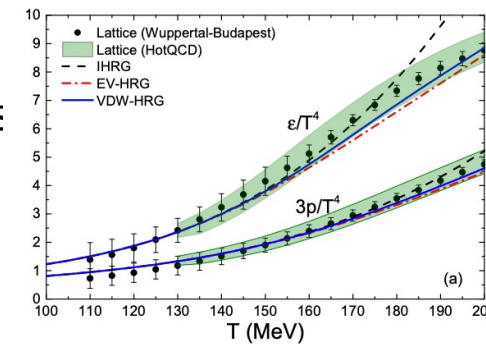
HRG model applications

- Heavy-ion collisions
 - Hadrochemistry (chemical freeze-out)
 - Fluctuations of conserved charges
- Lattice QCD context
 - Understanding the degrees of freedom
 - Equation of state, susceptibilities, partial pressures
- Early universe
 - Modeling QCD contribution to cosmic EoS
 - Finite isospin density
- Neutron-star matter
 - Extending to include non-resonant interactions
 - Hadronic part of the CMF model

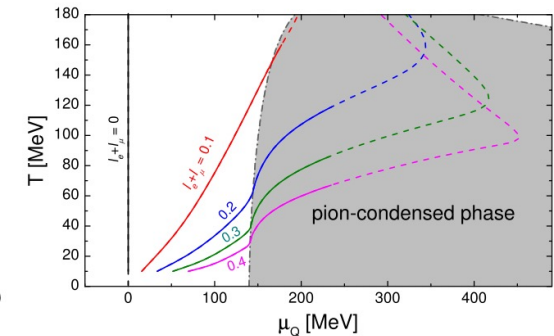
Natural block for MUSES



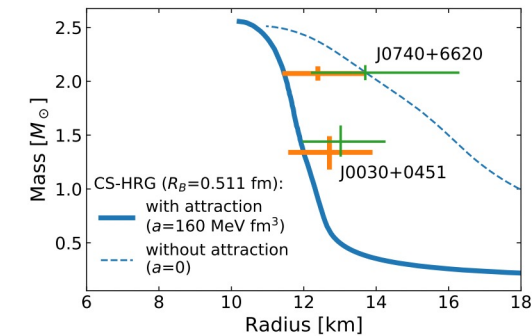
ALICE Coll. (2022)



VV et al., PRL (2017)



VV et al., PRL (2021)



Fujimoto et al., 2109.06799

HRG model aspects in Thermal-FIST

- Equation of state and related properties
 - thermodynamics, hadron yields and fluctuations
- Extensions of the base HRG model
 - finite resonance widths
 - repulsive (excluded volume) and van der Waals interactions (criticality)
 - (non-)conserved charges fluctuations and correlations
 - partial chemical equilibrium
- Heavy-ion applications
 - thermal fits
 - small systems and canonical effects
 - Monte Carlo event generator
 - partial chemical equilibrium
 - light nuclei
- Other applications
 - Neutron star matter
 - Early universe (cosmic EoS)

Thermal-FIST structure

- Core library (libThermalFIST)
 - Ideal (base) HRG model (HRGBase)
 - Interacting HRG model (HRGEV/HRGV DW)
 - Partial chemical equilibrium (HRGPCE)
 - Monte Carlo mode (HRGEventGenerator)
 - Thermal fits (HRGThermalFit)
- Graphical user interface (QtThermalFIST)
 - Based on Qt5
 - Wrapper around libThermalFIST
- Sample console applications
 - Essentially just C++ macros linking to libThermalFIST

External dependencies:

- Eigen library for linear algebra (header-only, built-in)
- Minuit2 (built-in, i.e. ROOT **not needed**)
- Qt6/Qt5 (for GUI only)

Using Thermal-FIST: Jupyter notebooks



Interactive notebooks through Jupyter (xeus kernel and ROOT-cling)*

A screenshot of a Jupyter notebook interface. The top bar shows the Jupyter logo, the notebook name 'FitExample', and a status message 'Last Checkpoint: несколько секунд назад (unsaved changes)'. On the right, there is a 'Logout' button and a 'Trusted' status indicator. Below the top bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', and 'Help'. A toolbar contains icons for file operations, cell navigation, and execution. The notebook content is divided into two sections. The first section, titled 'Initialize and run the fitter', contains a code cell with C++ code for setting chemical potentials, initializing the fitter, setting fit flags, and performing the fit. The second section, titled 'Print the fitted parameters and the χ^2 ', contains a code cell with C++ code for printing the fit results. The output of the second cell is displayed below the code, showing the extracted parameters: T [MeV] = 155.28 +/- 2.78665, R [fm] = 10.3342 +/- 0.545205, and chi2/dof = 15.3832/6.

```
jupyter FitExample Last Checkpoint: несколько секунд назад (unsaved changes) Logout
File Edit View Insert Cell Kernel Widgets Help Trusted C++14
[Icons] [Run] [Code]

Initialize and run the fitter

In [7]: // Set chemical potentials to zero
model.SetBaryonChemicalPotential(0.0);
model.SetElectricChemicalPotential(0.0);
model.SetStrangenessChemicalPotential(0.0);
model.SetCharmChemicalPotential(0.0);
model.FillChemicalPotentials();

// Initialize the fitter
ThermalModelFit fitter(&model);

// Do not fit muB, it is zero at LHC
fitter.SetParameterFitFlag("muB", false);

// Pass the data to the fitter
fitter.SetQuantities(dataPbPb010);

// Perform the fit
ThermalModelFitParameters fitResult = fitter.PerformFit(false);

Print the fitted parameters and the  $\chi^2$ 

In [8]: cout << "Extracted parameters:" << endl;
cout << setw(15) << "T [MeV]" << " = " << setw(15) << 1.e3 * fitResult.T.value << " +/- " << 1.e3 * fitResult.T.error << endl;
cout << setw(15) << "R [fm]" << " = " << setw(15) << fitResult.R.value << " +/- " << fitResult.R.error << endl;
cout << setw(15) << "chi2/dof" << " = " << setw(15) << fitResult.chi2 << "/" << fitResult.ndf << endl;

Extracted parameters:
      T [MeV] =      155.28 +/- 2.78665
       R [fm] =      10.3342 +/- 0.545205
    chi2/dof =      15.3832/6
```

*Since version 1.2.1, example at github.com/vlvovch/FIST-jupyter

Thermal-FIST and HRG model equation of state

Particle list file: `/Users/vlvovch/Code/Thermal-FIST/input/list/PDG2020/list-withnuclei.dat + decays.dat` Load particle list... Load decays...

Thermal model Thermal fits Equation of state Event generator Particle list editor

	Name	PDG ID	Mass	Stable?	Neutral?	B	Q	S	Prim. density	Prim. multiplicity	Total multiplicity	Scaled varian
1	pi0	111	0.134977	*	*				0.0461954	99.0734	302.421	1.12117
2	pi+	211	0.13957	*			+1		0.0456136	97.8257	264.084	1.11731
3	pi-	-211	0.13957	*			-1		0.0456136	97.8257	264.084	1.11731
4	f(0)(500)	9000221	0.475	2 decays	*				0	0	1.12992	1
5	K+	321	0.493677	*			+1	+1	0.0121721	26.1051	48.3981	1.01192
6	K-	-321	0.493677	*			-1	-1	0.0121721	26.1051	48.3981	1.01192
7	anti-K0	-311	0.497611	*				-1	0.0119642	25.6592	47.0889	1.01163
8	K0	311	0.497611	*				+1	0.0119642	25.6592	47.0889	1.01163
9	eta	221	0.547862									
10	rho(770)-	-213	0.77526									
11	rho(770)+	213	0.77526									
12	rho(770)0	113	0.77526									
13	omega(782)	223	0.78265									
14	K*(892)+	323	0.8955									
15	K*(892)-	-323	0.8955									
16	K*(892)0	313	0.89555									
17	anti-K*(892)0	-313	0.89555									
18	anti-p	-2212	0.938272									
19	p	2212	0.938272	*		+1	+1		0.00273362	5.86268	16.3447	0.993893
20	n	2112	0.939565	*		+1			0.00271554	5.82392	16.3059	0.993935
21	anti-n	-2112	0.939565	*		-1			0.00271554	5.82392	16.3059	0.993935

Correlations

FeedException: Final Type: Particle Quantity: Susceptibility

	pi0	pi+	pi-	K+	K-
pi0	0.420794	0.0939612	0.0939612	0.00733736	0.00733736
pi+	0.0939612	0.281041	0.107052	0.00201561	0.0123788
pi-	0.0939612	0.107052	0.281041	0.0123788	0.00201561
K+	0.00733736	0.00201561	0.0123788	0.0469004	0.00381837
K-	0.00733736	0.0123788	0.00201561	0.00381837	0.0469004

HRG model configuration:

Model: Quantum van der Waals Ensemble: Grand-canonical

Statistics: Boltzmann Quantum for All particles Use quadratures

Resonance widths: Zero-width EV/vdW parameter list...

Conservation laws... EV/vdW interactions... PCE/Saha/Other...

Parameters:

T (MeV): 155.00 V_B : 1.0000 V_S : 1.0000

μ_B (MeV): 0.00 μ_Q (MeV): 0.00 μ_S (MeV): 0.00

R (fm): 8.0000 R_C : 8.0000 V (fm³): 2144.66

B: 0 Q: 0 S: 0

Compute fluctuations and correlations ☒ Reset mu's

Calculate Calculate from fit tab Write to file...

ammas = 1 = 2144.66 fm³

particle density = 0.339627 fm⁻³

Net baryon density = 3.04932e-20 fm⁻³

Net baryon number = 6.53975e-17

Net electric charge = 1.45328e-17

Net strangeness = 0

Absolute baryon number = 98.8937

E/N = 0.918198

S/N = 6.90547

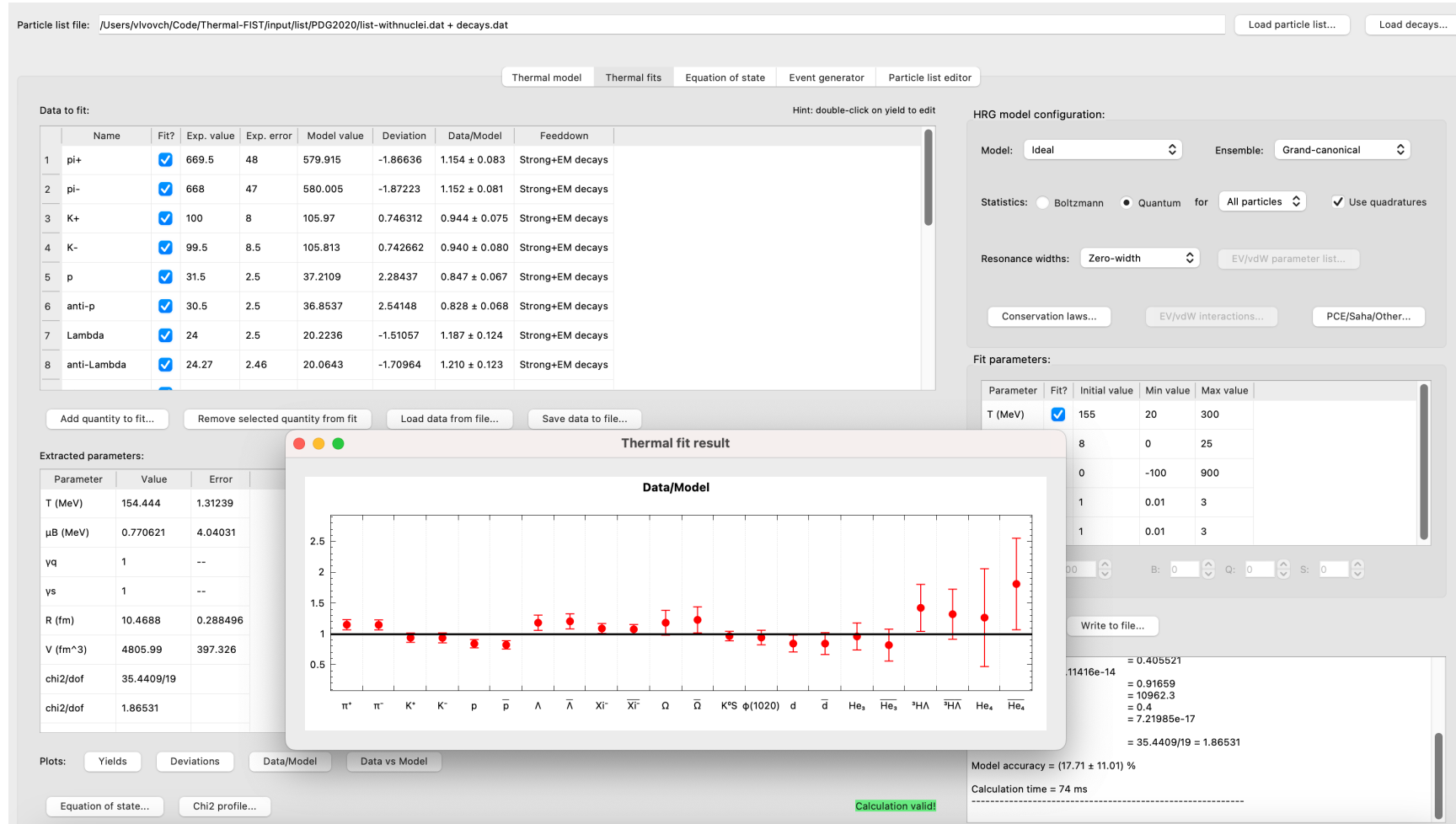
S/|S| = 0

Calculation time = 189 ms

Calculation valid!

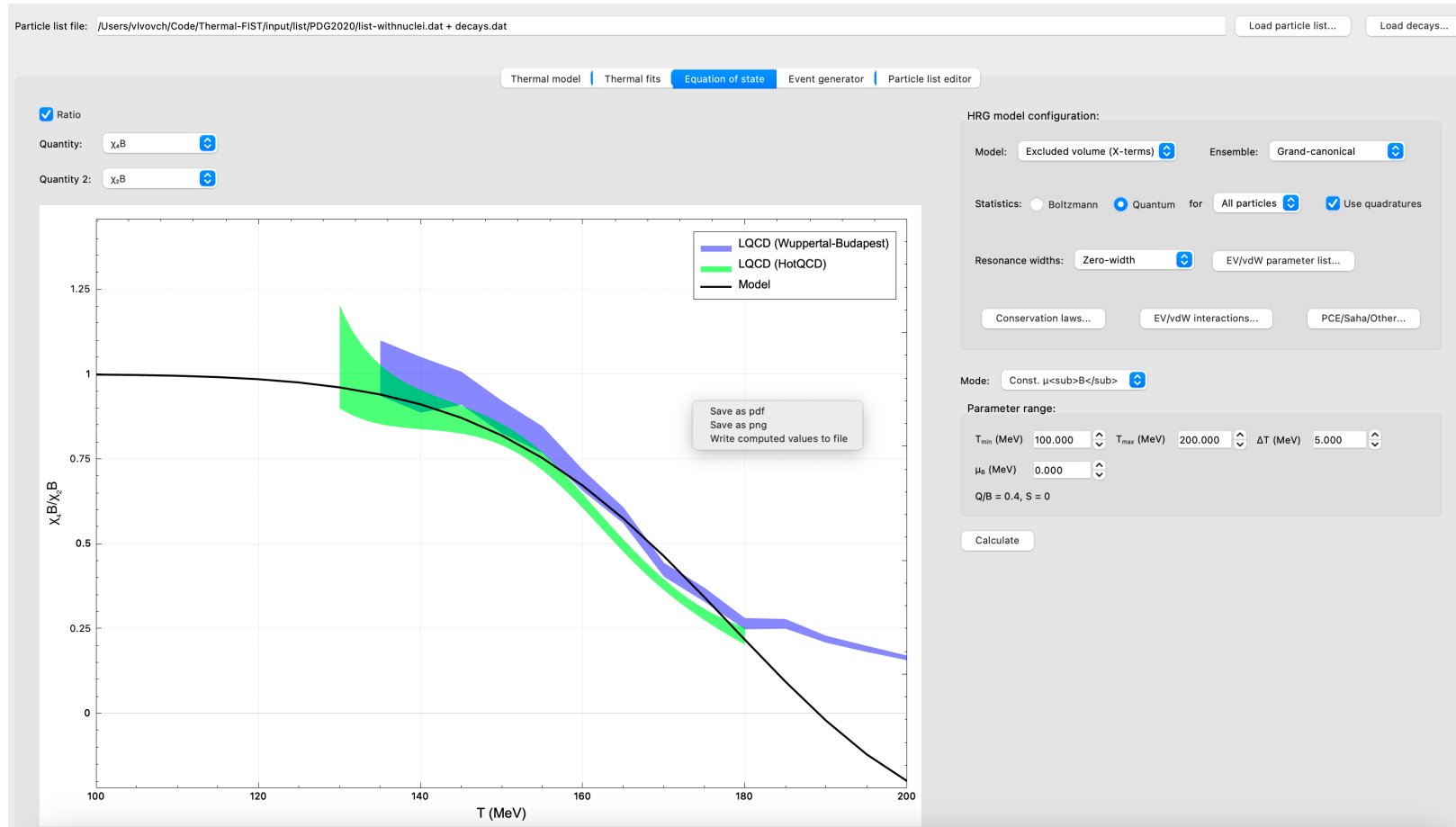
- Base calculation is at fixed T & $\mu_{B,Q,S}$ (alternatively at fixed s/n_B , Q/B , $S/|S|$)
- Thermodynamic functions, hadron abundances, feeddown, correlations and fluctuations

Thermal-FIST and Thermal-FITS



- Extract chemical freeze-out parameters from heavy-ion hadron abundances
- χ^2 minimization

Thermal-FIST and equation of state



- Compute HRG model quantities along a fixed T , μ_B , or μ_B / T
- Impose conservation laws [e.g. strangeness neutrality (heavy-ions) or charge neutrality (neutron stars)]

Thermal-FIST and equation of state

Console mode provides more flexibility

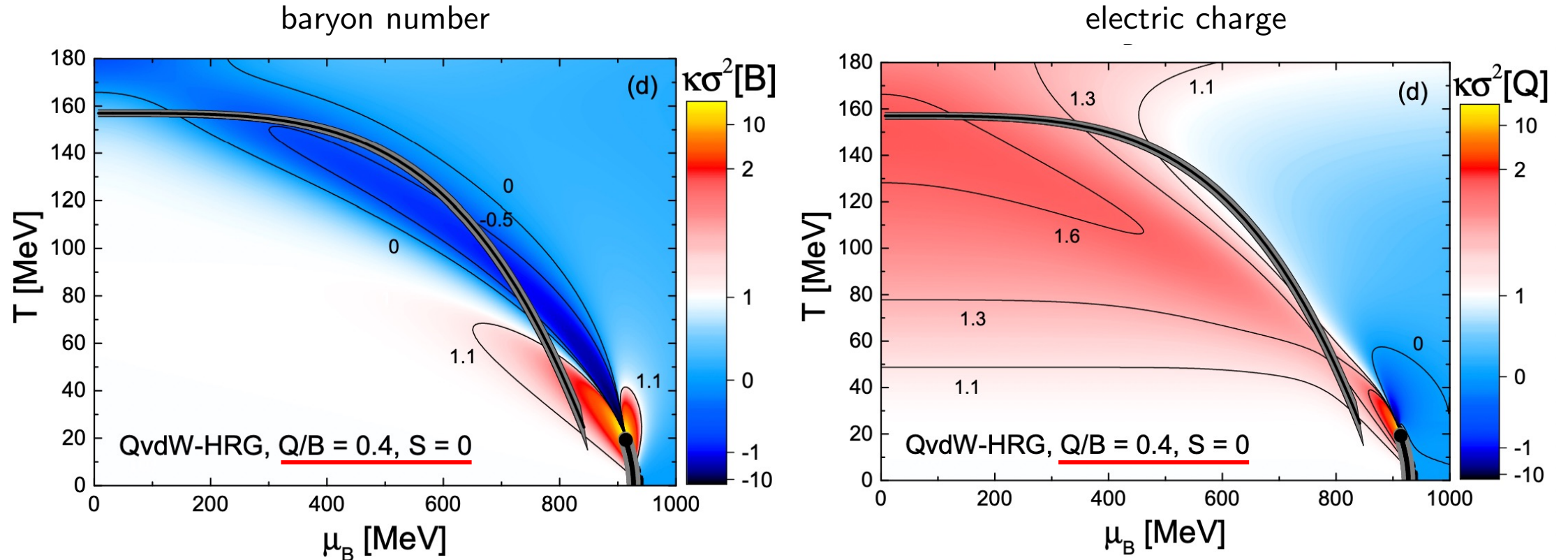
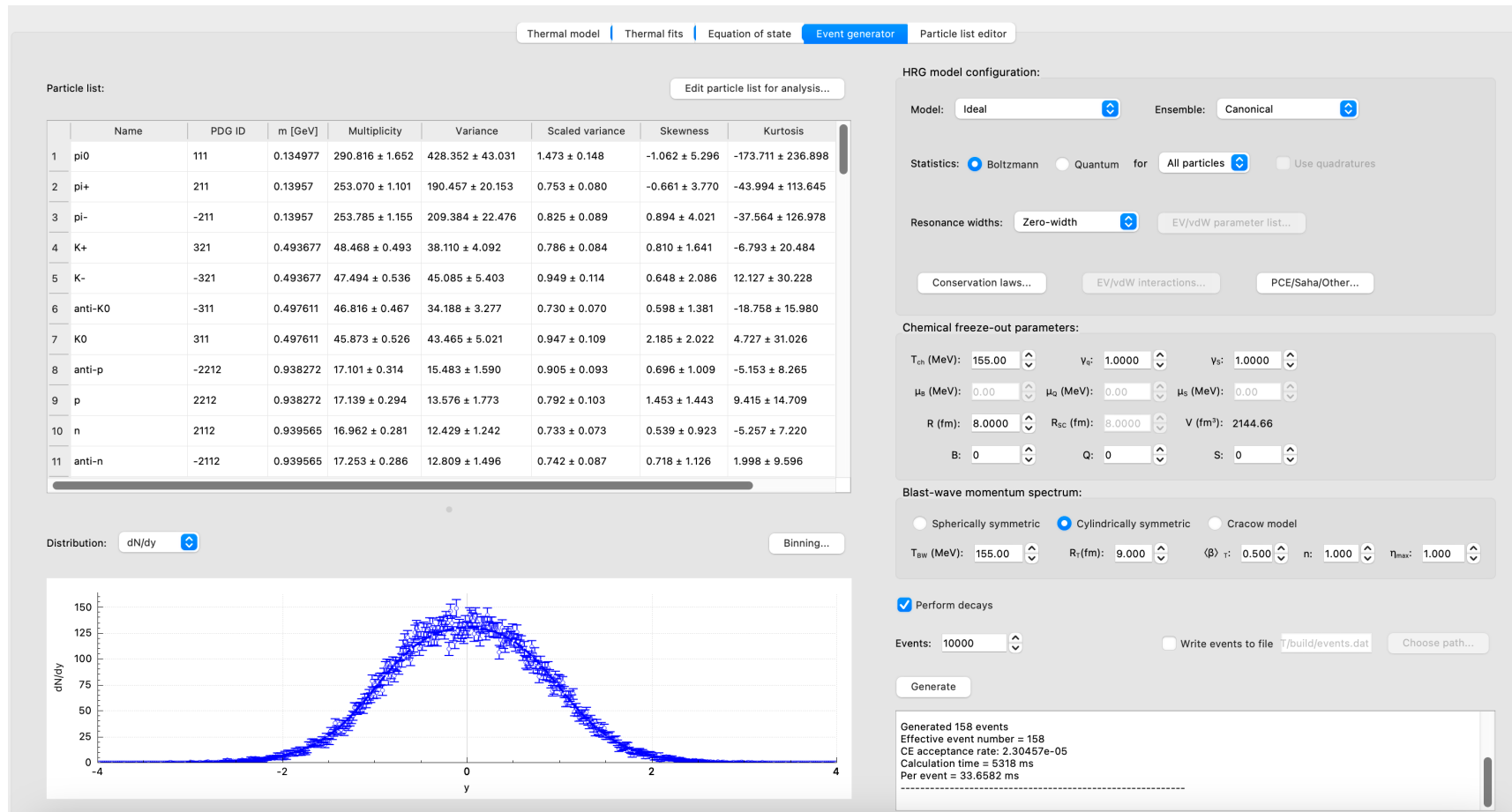


Figure from R.V. Poberezhnyuk et al., PRC 99, 024907 (2019)

Thermal-FIST and HRG event generator



- Monte Carlo sampling of hadron abundances, momenta, and coordinates
- Superimposed on blast-wave flow velocity profile
- Realistic modeling of acceptance effects, especially for correlations and fluctuations

Interactions in HRG

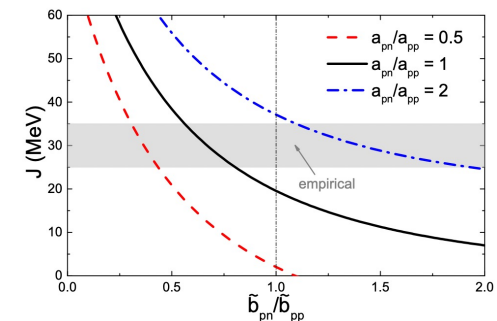
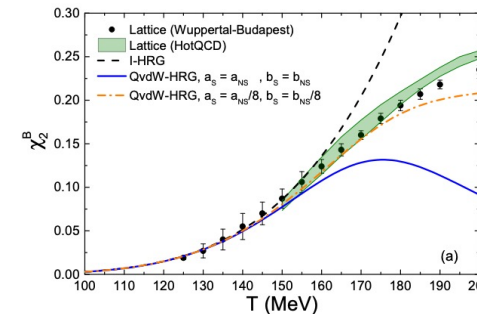
Interacting HRG can be more smoothly connected to other QCD phases than ideal HRG

e.g. the crossover in T direction, Albright, Kapusta, Young, PRC (2014)

Thermal-FIST incorporates van der Waals interactions in the most general form

$$p(T, n_1, \dots, n_h) = \sum_i \frac{T n_i}{1 - \sum_j \tilde{b}_{ji} n_j} - \sum_{i,j} a_{ij} n_i n_j$$

- Separate excluded volume b_{ij} for each pair i,j of species
- Separate mean field a_{ij} for each pair i,j of species
- So far very little explored!
- E.g. indications for flavor-dependent parameters from
 - Lattice QCD susceptibilities [Karthein et al., 2107.00588]
 - Symmetry energy
 - Neutron-star matter EoS [Fujimoto et al., 2109.06799]



VV, Motornenko, Alba, et al., 1707.09215

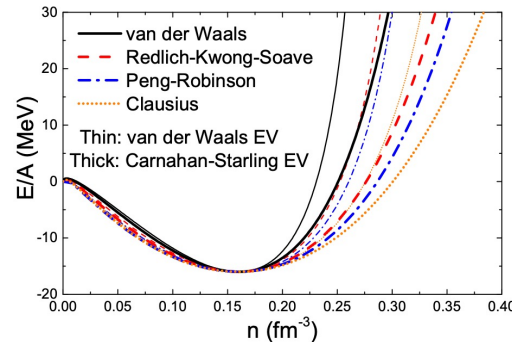
Interactions in HRG: Beyond van der Waals

Standard van der Waals gives too stiff EoS beyond the saturation density

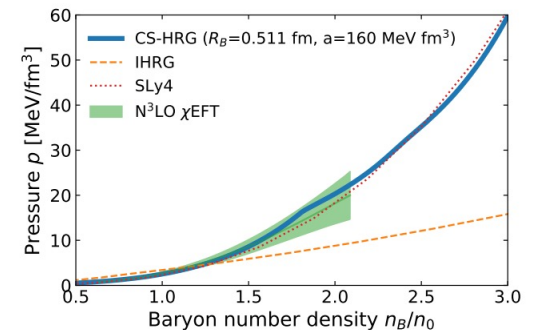
Beyond vdW:

- Generalized (non-linear) excluded volume
 - Carnahan-Starling (CS)
- Density-dependent mean-field
 - Real gases
 - Skyrme
 - VDF model

A. Sorensen, V. Koch, PRC 104, 034904 (2021)



VV, PRC 96, 015206 (2017)

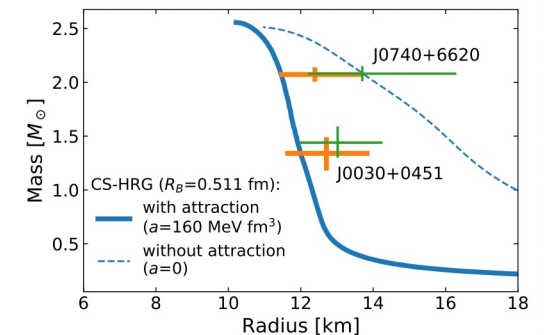


Helps soften the EoS in the cold & dense regime, making it easier to match with others

Available out-of-the-box in FIST in next version (already present in **devel** branch on github), adding leptons into the list one can do neutron-star matter

Another extension: pion interactions and condensation at finite isospin density

VV et al., PRL (2021)



Fujimoto et al., 2109.06799